

# Quantum Fault Tolerance Meets Toom's Contours

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Change letters: the  $d$  in the model section refers to a size that is comparable to the code distance where here  $d$  refers to the dimension of the Toric complex

## 1 The $(d, d')$ -Dimensional Toric Code and the SWEEP Rule

**Definition 1.1** ( $d$ -Dimensional Toric Complex). For integers  $d$  and  $n$ , denote the group resulting from the  $d$ -directed power of the cyclic group of order  $n$  by  $\mathcal{G}_n^d = \mathbb{Z}_n^{\times d}$ , and denote the standard generating set of  $\mathcal{G}_n^d$  by  $S = \{1_i : i \in [d]\}$ . We define the  $d$ -dimensional toric complex to be the hypergraph obtained from the Cayley graph  $\text{Cay}(\mathcal{G}_n^d, S)$  by taking its vertices as the 0-faces and for any  $i \in [d-1]$  we define the  $i$ -faces as follow: First, for a vertex  $v \in \mathcal{G}_n^d$  and for a subset of generators  $S' \subset S$  of size  $|S'| = i$ , the  $i$ -face rooted at  $v$  and spanned by  $S'$ , denoted by  $\theta_{v,S'}$ , is:

$$\theta_{v,S'} := \bigcup_{I \subset S'} \left\{ \left( \prod_{g \in I} g \right) v \right\}$$

The  $i$ -faces of the complex are the collection of all the possible rooted spanned  $i$ -faces induced by  $\text{Cay}(\mathcal{G}_n^d)$ . We will denote them by  $\mathcal{F}_i$ .

Additionally, we name  $n$  and  $d$ , the *side length* and the *dimension* of the complex.

**Definition 1.2** (Positive Edges and Cubes). For  $u, v \in \mathcal{G}_n^d$ , and  $S'_1, S'_2 \subset [d]$ , with sizes  $|S'_1| = |S'_2| = i$ , we say that the  $i$ -faces  $\theta_{v,S'_1}$  and  $\theta_{u,S'_2}$  are adjacent if the intersection  $\theta_{v,S'_1} \cap \theta_{u,S'_2}$  is an  $i-1$ -face. In addition, if  $\theta_1$  and  $\theta_2$  are  $i$  and  $i-1$ -faces such that  $\theta_2 \subset \theta_1$ , then we say that  $\theta_2$  is a *positive edge relative to face*  $\theta_1$  if they are rooted at the same vertex. Furthermore, we say that  $\theta_1$  is a *positive cube* of  $\theta_2$ . When the  $i$ -face is clear from the context, we will abbreviate and say that  $\theta_2$  is *positive*.

**Example 1.3.** For example, let  $v, u \in \mathcal{G}_n^d$  and  $g_1 \neq g_2 \in S$ . Consider the 2-face  $\theta = \{v, g_1v, g_2g_1v, g_2v\}$ . Then, for  $x, y \in \mathcal{G}_n^d$ , we say that an edge (1-face)  $\{x, y\}$  is *positive relative to the face*  $\theta$  if  $x = v$  and  $y$  is either  $g_1v$  or  $g_2v$ .

We denote the complex by  $G = (V, E, F)$ , where  $V$ ,  $E$ , and  $F$  are the 0, 1, and 2 faces.

Add a paragraph that explains that we choose not to use a chain-complex since we earn nothing from it i.e don't care about topology properties. Yet mention that chain-complex formulation proved itself as an effective tool to compute the properties of the codes we use.

**Definition 1.4** (Assignment on The  $(d', d)$ -Toric Complex). Let  $\mathcal{T}$  be a  $d$ -dimensional complex. Each of its  $i$ -face sets naturally induces a vector space  $\mathbb{F}_2^{|\mathcal{F}_i|}$ . We call a binary vector  $z \in \mathbb{F}_2^{|\mathcal{F}_i|}$  an assignment on the  $i$ -faces of  $\mathcal{T}$ . For a vertex  $v$  and a subset of generators  $S' \subset S$  of size  $|S'| = i$ , we use the notation  $\theta_{\mathbf{v}, S'}$  to represent the unit vector in  $\mathbb{F}_2^{|\mathcal{F}_i|}$  that has 0 in every coordinate except the one associated with the  $i$ -face  $\theta_{v, S'}$ . Then

$$\left\{ \theta_{\mathbf{v}, S'} : v \in \mathcal{F}_0, S' \subset S \text{ s.t } |S'| = i \right\}$$

is a base to  $\mathbb{F}_2^{|\mathcal{F}_i|}$  **Not exactly base, they are linear dependent.** Let  $z$  be an assignment on the  $i$ -faces. Let  $v_1, \dots, v_k \in \mathcal{F}_0$  and  $S_1, \dots, S_k \subset S^{\times k}$  be the generator subsets such that  $\theta_{v_1, S'_1}, \theta_{v_2, S'_2}, \dots, \theta_{v_k, S'_k}$  are the collection of  $i$ -faces that support  $z$ , namely:

$$z = \sum_{i=1}^k \theta_{\mathbf{v}_i, S'_i}$$

We refer to  $\theta_{v_1, S'_1}, \theta_{v_2, S'_2}, \dots, \theta_{v_k, S'_k}$  as the collection of faces that form  $z$ , or briefly, the faces form  $z$ . For  $i > 0$ , we define the linear map  $\partial : \mathbb{F}_2^{|\mathcal{F}_i|} \rightarrow \mathbb{F}_2^{|\mathcal{F}_{i-1}|}$  such that

$$\partial \theta_{\mathbf{v}, S'} := \sum_{g \in S'} \theta_{\mathbf{v}, S'/g} + \sum_{g \in S'} \theta_{\mathbf{g}\mathbf{v}, S'/g}$$

For  $i < d$ , we define the linear map  $\partial^* : \mathbb{F}_2^{|\mathcal{F}_i|} \rightarrow \mathbb{F}_2^{|\mathcal{F}_{i+1}|}$  such that

$$\partial^* \theta_{\mathbf{v}, S'} := \sum_{g \in S/S'} \theta_{\mathbf{v}, S' \cup g} + \sum_{g \in S'} \theta_{\mathbf{g}^{-1}\mathbf{v}, S' \cup g}$$

We define the *restriction of  $z$  to a vertex  $v$*  to be the linear map  $|_v : \mathbb{F}_2^{|\mathcal{F}_i|} \rightarrow \mathbb{F}_2^{|\mathcal{F}_i|}$  such that:

$$\theta_{\mathbf{u}, S'}|_v := \begin{cases} \theta_{\mathbf{u}, S'} & v \in \theta_{\mathbf{u}, S'} \\ 0 & \text{else} \end{cases}$$

We define the *front of  $z$  to a vertex  $v$*  to be the linear map  $|_{v+} : \mathbb{F}_2^{|\mathcal{F}_i|} \rightarrow \mathbb{F}_2^{|\mathcal{F}_i|}$  such that:

$$\theta_{\mathbf{u}, S'}|_{v+} := \begin{cases} \theta_{\mathbf{u}, S'} & u = v \\ 0 & \text{else} \end{cases}$$

We say that an assignment  $z$  is *connected* if the graph that is induced by taking the faces that support  $z$  as vertices and defining the edges to be the adjacency relation between them is connected. We define the connected components of  $z$  to be its subsets such that each of them is a connected assignment. For a  $j$ -face  $f$ , we say that  $f \in z$  if  $z$  has a support on  $\theta_{\mathbf{v}, \mathbf{S}'}$  such that  $f \in \theta_{\mathbf{v}, \mathbf{S}'}$ . Consider an assignment  $z \in \mathbb{F}_2^{\mathcal{F}_i}$  and a vertex  $v \in \mathcal{G}_n^d$ . We say that  $z$  is a *rooted assignment* at vertex  $v$  if there are subsets  $S'_1, \dots, S'_k \subset S^{\times k}$  such that each of  $S'_j$  is at size  $i$  and  $z$  is decomposed as:

$$z = \sum_j^k \theta_{\mathbf{v}, \mathbf{S}'_j}$$

Put differently,  $z$  is the result of taking a collection of  $i$ -faces that are rooted at  $v$ .

Add a proof for  $\partial\theta_{\mathbf{v}, \mathbf{S}'} \cdot \partial^*\theta_{\mathbf{u}, \mathbf{S}''} = 0 \Rightarrow$  gives a CSS code.

Need to decide on what should I elaborate, for example should I provide a proof for the distance and the rate of the code? [Den+02]

**Definition 1.5** ( $(d, d')$  Toric Code). Let  $d, d'$  be integers such that  $1 < d' < d$ . For any integer  $n$ , we construct the  $n$ th instance of the quantum code family  $(d, d')$  *Toric Code* as follows: Let  $G = (\mathcal{F}_0, \dots, \mathcal{F}_d)$  be the  $d$ -dimensional toric complex obtained from  $\mathcal{G}_n^d$  as defined in Definition 1.1. We associate the (q)bits with the  $\mathcal{F}_{d'}$ , the  $X$ -checks with the  $\mathcal{F}_{d'-1}$ , and the  $Z$ -checks with the  $\mathcal{F}_{d'+1}$ . An  $X$ -check,  $c_x$ , is satisfied if the parity of (q)bits set on the  $d'$ -face containing  $c_x$  is even. Similarly, a  $Z$ -check,  $c_z$ , is satisfied if the parity, in the Hadamard basis, of the (q)bits set on the  $d'$ -faces contained in  $c_z$  is even.

**Definition 1.6** (Sweep Rule). Let  $z$  be an assignment over the  $d'$ -faces, The *sweep rule decoding procedure* at vertex  $v \in \mathcal{G}_n^d$ , defining as follow:

1. For the  $\mathcal{C}_X$  code. Look for a rooted assignment  $\tilde{z}$  at  $v$  such that  $(\partial\tilde{z})|_{v+} = \partial(z)|_{v+}$ . Namely, the  $X$ -checks rooted at  $v$  get the same syndromes for  $z$  and  $\tilde{z}$ . If there is one, then flip  $\tilde{z}$ .
2. Apply the Hadamard gate on any  $d'$  face containing  $v$ .
3. For the  $\mathcal{C}_Z$  code. Look for a rooted assignment  $\tilde{z}$  at  $v$  such that  $(\partial^*\tilde{z})|_{v+} = \partial^*(z)|_{v+}$ . Namely, the  $Z$ -checks rooted at  $v$  get the same syndromes for  $z$  and  $\tilde{z}$ . If there is one, then flip  $\tilde{z}$ .
4. Apply the Hadamard gate again on any  $d'$  face containing  $v$  to reverse the second stage.

Stages 1 and 3 involve measurement of stabilizers.

**Definition 1.7** (Decoding Gadget). Consider a quantum circuit  $C$ , for a quantum gate  $u$  in  $C$ , with fun-in  $\Delta$ , we say that  $u$  is a *decoding gadget* if it has the following form:

1. The wires of  $u$  wires can be decomposed into two subsets  $A$  and  $B$  such that the wires in  $A$  are reset wires, and there exists an encoded register  $R$  in  $C$  such that the wires in  $B$  are all set in  $R$ .
2. For a set of faults  $\mathcal{E}$ , there is a Pauli gate  $u'$  that acts only on  $B$  such that the action of  $u$  in  $C[\mathcal{E}]$  equals  $u'$ . Put differently, the quantum circuit obtained from  $C[\mathcal{E}]$  by replacing  $u$  with  $u'$  equals  $C[\mathcal{E}]$ .

**Definition 1.8** (Unitary Decoding Representation). Let  $C = \{u_i\}_{i=1}$  be a quantum circuit, and let  $\mathcal{U} = \{u'_i\}_{i=1} \subset C$  be a subset of the gates in  $C$  that are decoding gadgets. For a set of faults  $\mathcal{E}$ , we define the *unitary decoding representation* of  $C[\mathcal{E}]$  to be the quantum circuit obtained by replacing each of the gates in  $\mathcal{U}$  with their corresponding Pauli gate according to  $\mathcal{E}$ .

**Claim 1.9.** The local gates performing Toom's rule are decoding gadgets.

*Proof.* [prove that](#). □

**Definition 1.10** (Registers Coordinates System). Let  $C = \{u_i\}_{i=1}$  be a quantum circuit, with width  $w$  and registers  $\mathcal{R} = R_1, \dots, R_k$ , each of length at most  $n$ . For register  $R_i$ , we define the register identifier  $\mathbf{1}_{R_i} = \mathbf{1}_i$ . Then, we define the map  $\phi_{\mathcal{R}}: \mathbb{Z}_w \rightarrow \mathbb{Z}_R \times \mathbb{Z}_n$  to return, for a space-location coordinate  $x \in \mathbb{Z}_w$ , the tuple consisting of the identifier of the register containing  $x$ , and its relative coordinate within the register. Namely,

$$\psi_{\mathcal{R}}(x) = (\mathbf{1}_{R_j}, x')$$

Where  $R_j$  is the register containing  $x$ , and  $x'$  is the index of the qubit, set at the  $x$  location, in  $R_j$ . The presentation of  $C$  in the *registers coordinate system* is the encoding by a set of gates  $\{u'_j\}_{j=1}$ , which is the result of replacing each space-location of the original gates with the applications of  $\psi_{\mathcal{R}}$  on them. Namely, for each  $u_i = (g_i, l_i) \in C$ , we map its space-location coordinate  $l_{i_2} = (l_{i_2}^{(1)}, \dots, l_{i_2}^{(\Delta)})$  to:

$$l_{i_2} \mapsto (\psi_{\mathcal{R}}(l_{i_2}^{(1)}), \dots, \psi_{\mathcal{R}}(l_{i_2}^{(\Delta)}))$$

For brevity, we name the first and second coordinates in the result of  $\psi_{\mathcal{R}}$  the *register coordinate* and the *inner coordinate*.

**Definition 1.11** (Base graph  $G_o$ ). Let  $C$  be a quantum circuit, at depth  $d$ , with registers  $R_1, \dots, R_k$ , all of which are encoded by Toric codes with the same parameters (i.e., dimension, length, etc.). Let's denote by  $\mathcal{G}_1, \dots, \mathcal{G}_k$  the Toric complexes that induce the codes. We define the base graph of  $C$ , denoted by  $G_o = (V_o, E_o)$  as follows:

1. The vertices  $V_o$  are tuples, where the first coordinates correspond to a vertex in the union over the vertices in  $\mathcal{G}_1, \dots, \mathcal{G}_k$ , and the second coordinate is associated with a time. Namely:

$$V_o := \{(v, t) : t \in [d], v \in \bigcup_i \mathcal{G}_i\}$$

2. The edges  $E_o$  are derived from the original edges in  $\mathcal{G}_1, \dots, \mathcal{G}_k$ , with the addition that any two vertices in preceding times  $t$  and  $t + 1$  that support qubits (faces) such that  $C$  acts non-trivially on those qubits at time  $t$  are also connected by an edge. Namely:

$$E_o := \{ \{(v, t), (u, t')\} : \text{if} \\ \text{either there exists } i \text{ such } \{v, u\} \in \mathcal{G}_i \text{ and } t' = t, t + 1 \\ \text{or there exists } u_i = (g_i, l_i) \in C \text{ such } v, u \in l_{i,2} \text{ and } t = l_{i,1}, t' = l_{i,1} + 1 \}$$

## 2 Infinite Torics

**Definition 2.1** (Shifting in Registers System). Consider the register coordinate system induced by a quantum circuit of width  $w$ , with  $R$  registers, each of length at most  $n$ . For an integer  $r \in \mathbb{Z}$ , we define the  $r$ -shifting function  $\text{Sh}_r : (\mathbb{Z}_R \times \mathbb{Z}_n)^{\times \Delta} \rightarrow (\mathbb{Z}_R \times \mathbb{Z})^{\times \Delta}$  to act on space-location coordinates as follows:

$$\text{Sh}_r(l_1, \dots, l_k) = (\text{Sh}_r(l_1), \dots, \text{Sh}_r(l_k))$$

$$\text{Where } \text{Sh}_r(l_i) = \begin{cases} l_i + r & \text{if } l_i \text{ is an inner coordinate} \\ l_i & \text{else} \end{cases}$$

**Definition 2.2.** Let  $n$  be an integer, and let  $C = \{u_i\}_{i=1}$  be a quantum circuit with registers  $\mathcal{R} = R_1, \dots, R_k$  such that each register  $R_i$  is a toric encoding with a side length  $n$ . In addition, all the decoding gadgets of  $C$  are sweep rule gadgets; let us denote them by  $\mathcal{U}$ . We define the  $\infty$ -extension of  $C$ , denoted by  $C^\infty$ , as follows:

1. For a register  $R_i \in \mathcal{R}$ , denote by  $d_i, d'_i$  the dimension of the toric complex that induces the code, and the dimension of the faces that are associated with the bits. Each register  $R_i$  in  $\mathcal{R}$  is mapped to a register  $R'_i$ , which is a  $(d_i, d'_i)$ -dimensional, infinite side length, toric encoding. Since the mapping between registers is one-to-one, we use the identifier  $\mathbf{1}_{R_i}$  to identify also  $R'_i$ .
2. Each gate  $u_i \in C/\mathcal{U}$  is mapped to:

$$u_i = (g_i, l_i) \mapsto \bigcup_{j=-\infty}^{\infty} \{(g_i, (\text{Sh}_{jn}(l_i)))\}$$

3. Each gate  $u \in \mathcal{U}$  is mapped to the collection of sweep rule gadgets  $\{v_j\}_{j=-\infty}^{\infty}$  such that each of them has the same time coordinate as  $u$ , is in the register that corresponds to the image of  $u$ 's register, and is rooted at  $\text{root}(v) + jn$ .

For a set of faults  $\mathcal{E} = \{f_i\}_{i=1}$ , we denote by  $C^\infty[\mathcal{E}]$  the  $\infty$ -extension of  $C[\mathcal{E}]$ .

**Claim 2.3.** Let  $u = (g, l)$  and  $v_0, v_{\pm 1}$  plantation of  $u$  at locations  $\text{root}(u)$  and  $\text{root}(u) \pm n$ . Then  $v_0 v_{\pm 1}|_{[w]} = u$ .

*Proof.* [Prove it](#) □

**Definition 2.4.** For a quantum circuit  $C$ , of the form defined in Definition 2.2, and a set of faults  $\mathcal{E}$ , denote:

$$\begin{aligned}\rho &:= C[\mathcal{E}] |0\rangle\langle 0| C[\mathcal{E}]^\dagger \\ \rho_\infty &:= C^\infty[\mathcal{E}^\infty] |0\rangle\langle 0| (C^\infty[\mathcal{E}^\infty])^\dagger\end{aligned}$$

We say that a quantum circuit  $C$  is  $\infty$ -compatible if, for any set of faults  $\mathcal{E} = \{f_i\}_{i=1}$ , it holds that:

$$\rho = \mathbf{Tr}_{/[w]}(\rho_\infty)$$

**Claim 2.5.** Let  $C$  be quantum circuit of the form defined in Definition 2.2. Then  $C$  is  $\infty$ -compatible.

*Proof.* Its enough to prove the claim for circuits in their unitary decoding representation. We prove it by induction on the number of the gates.

1. Base. For an empty circuit,  $\rho_\infty$  is just the infinite string of zeros; thus, the restriction of it to  $[w]$  locations, namely  $\mathbf{Tr}_{/[w]}(\rho_\infty)$ , yields a single matrix element corresponds to the  $w$ -length string of zeros, which is exactly  $\rho$ .
2. Step. Assume the correction of the claim for any circuit, with less than  $\tau$  steps, and consider a circuit  $C$  with  $\tau$  steps. Denote by  $C'$  the circuit obtained from  $C$  by erasing the last gate in  $C$ , observes that  $C'$  is a circuit with  $\tau - 1$  steps. Denote by  $\rho'$  and  $\rho'_\infty$  the: [Enter it](#). By the induction assumption we have:

$$\rho' = \mathbf{Tr}_{/[w]}(\rho'_\infty)$$

Denote the last gate in  $C$  by  $u$ , and by  $I = \{v_j\}_{j=-\infty}^\infty$  the collection of gates in  $C^\infty$  to which  $u$  is mapped. Let us split into cases, depending on whether  $u$  is or is not a decoding gate.

- (a) Suppose that  $u$  is not a decoding gate. Let  $g$  and  $l$  be the gate element and the location of  $u$ . By the definition of  $\infty$ -extension,  $I$  equals:

$$I = \bigcup_{j=-\infty}^\infty \{(g, (\text{Sh}_{jn}(l)))\}$$

So there is only one gate in  $I$ , corresponding to  $j = 0$ , that has support on qubits with spatial location in the range  $[w]$ . Denote that gate by  $v_0$ .

$$\begin{aligned}
\mathbf{Tr}_{/[w]}(\rho_\infty) &= \mathbf{Tr}_{/[w]} \left( \prod_{j=-\infty}^{\infty} v_j \rho'_\infty \prod_{j=-\infty}^{\infty} v_j^\dagger \right) \\
&= v_0 \mathbf{Tr}_{/[w]} \left( \prod_{j=-\infty, j \neq 0}^{\infty} v_j \rho'_\infty \prod_{j=-\infty, j \neq 0}^{\infty} v_j^\dagger \right) v_0^\dagger \\
&= v_0 \mathbf{Tr}_{/[w]} \left( \prod_{j=-\infty, j \neq 0}^{\infty} v_j^\dagger \prod_{j=-\infty, j \neq 0}^{\infty} v_j \rho'_\infty \right) v_0^\dagger \\
&= v_0 \mathbf{Tr}_{/[w]}(\rho'_\infty) v_0^\dagger \\
&= v_0 \rho' v_0^\dagger \\
&= \rho
\end{aligned}$$

- (b) In the case that  $u$  is a decoding gate, observe that there are at most two gates in  $I$  that have support on the qubits with space-location in  $[w]$ . If there is exactly one, then repeating the non-decoding gate case while setting this gate as  $v_0$  gives the desired result. Else, in the case where two gates act non-trivially on qubits at  $[w]$ , denote those gates by  $v_0, v_1$ . Since we assumed that  $C$  is given in its unitary decoding representation, it holds that  $v_0, v_1$  are Paulis, in particular a product operator. Denote their product decompositions by  $v_0 = \prod_k v_0^{(k)}$  and  $v_1 = \prod_k v_1^{(k)}$ . Thus, by repeating the non-decoding case, but extracting from the infinite product only the terms in  $v_0$  and  $v_1$  that act non-trivially at  $[w]$ , we get:

$$\mathbf{Tr}_{/[w]}(\rho_\infty) = v_0 v_1|_{[w]} \rho' (v_0 v_1|_{[w]})^\dagger$$

Moreover, since  $v_0$  and  $v_1$  are a plantation of  $g$  in locations  $\text{root}(u)$  and  $\text{root}(u) \pm n$ , then by Claim 2.3, we have that  $u = v_0 v_1|_{[w]}$ ; therefore:

$$\begin{aligned}
\mathbf{Tr}_{/[w]}(\rho_\infty) &= v_0 v_1|_{[w]} \rho' (v_0 v_1|_{[w]})^\dagger \\
&= u \rho' u^\dagger \\
&= \rho
\end{aligned}$$

□

Add that the computation graph is lifted naturally to the  $\infty$ -extension, since the image of a mapping are overlapped gates, we can execute them simultaneously. Same for the base graph  $G_o \rightarrow G_o^\infty$ .